

8. Introduction to Seismic Processing TPG 4190

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Learning goals

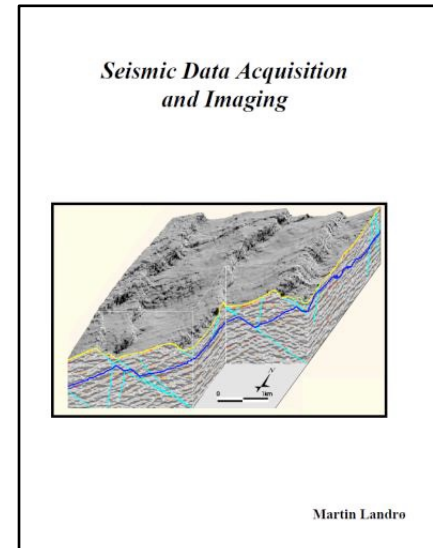
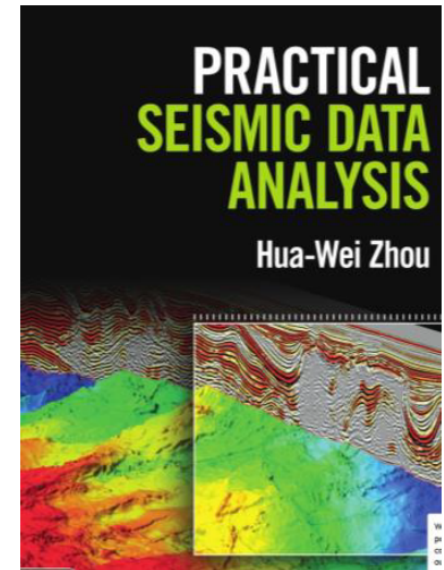
- This week's lectures are designed to facilitate a smooth start to the processing practicals;
- This short lecture revises the key steps in, and terminology of, marine 2D seismic data processing, with particular emphasis on the role of Common Midpoint (CMP) sorting;
- It also seeks to explain the overall aims of seismic data processing;
- After this lecture the students should understand the purpose of the main processing steps they are to apply in SeisSpace.

Recommended Reading

- For this introduction and preprocessing:

Zhou, *Practical Seismic Data Analysis*:
Chapter 2

Landrø, *Seismic Data Acquisition and Imaging*:
Sections 3.1-3.4 and 3.8

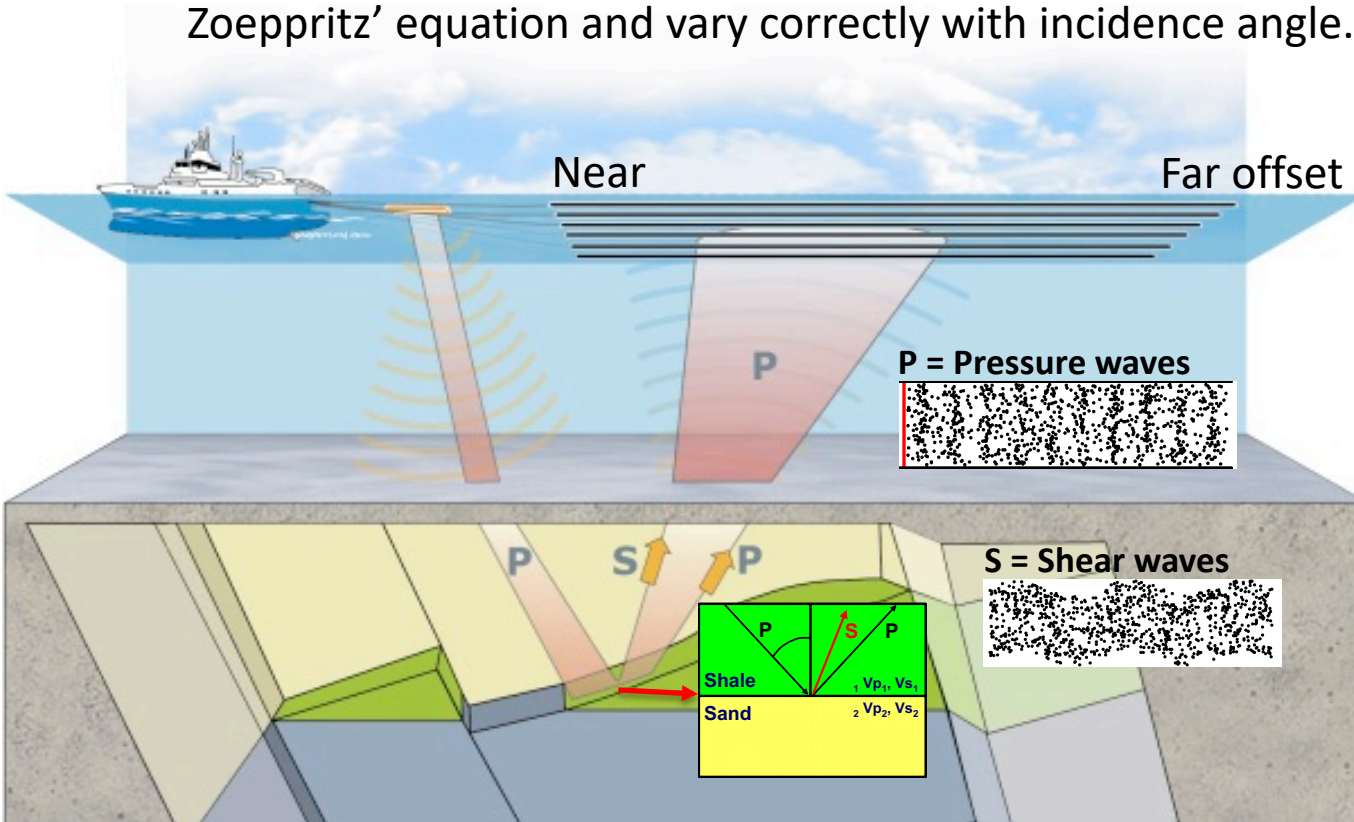


Outline

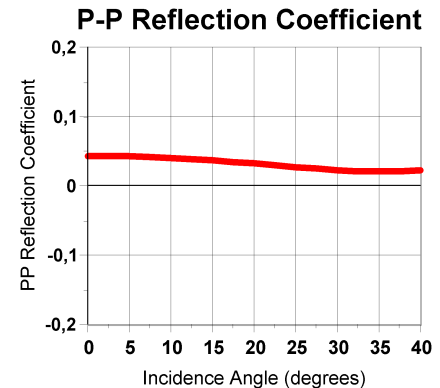
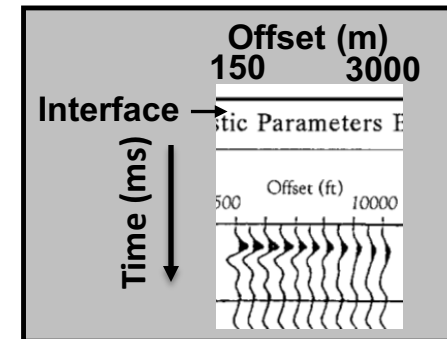
- What is the aim of seismic processing?
- Signal and noise
- A simple 2D processing flow illustrated
 - Shots and muting
 - Common midpoint gathers
 - Normal moveout and RMS velocities
 - Velocity analysis and stack
 - Removing multiples
 - Imaging
- Summary

The aims of reflection seismic processing

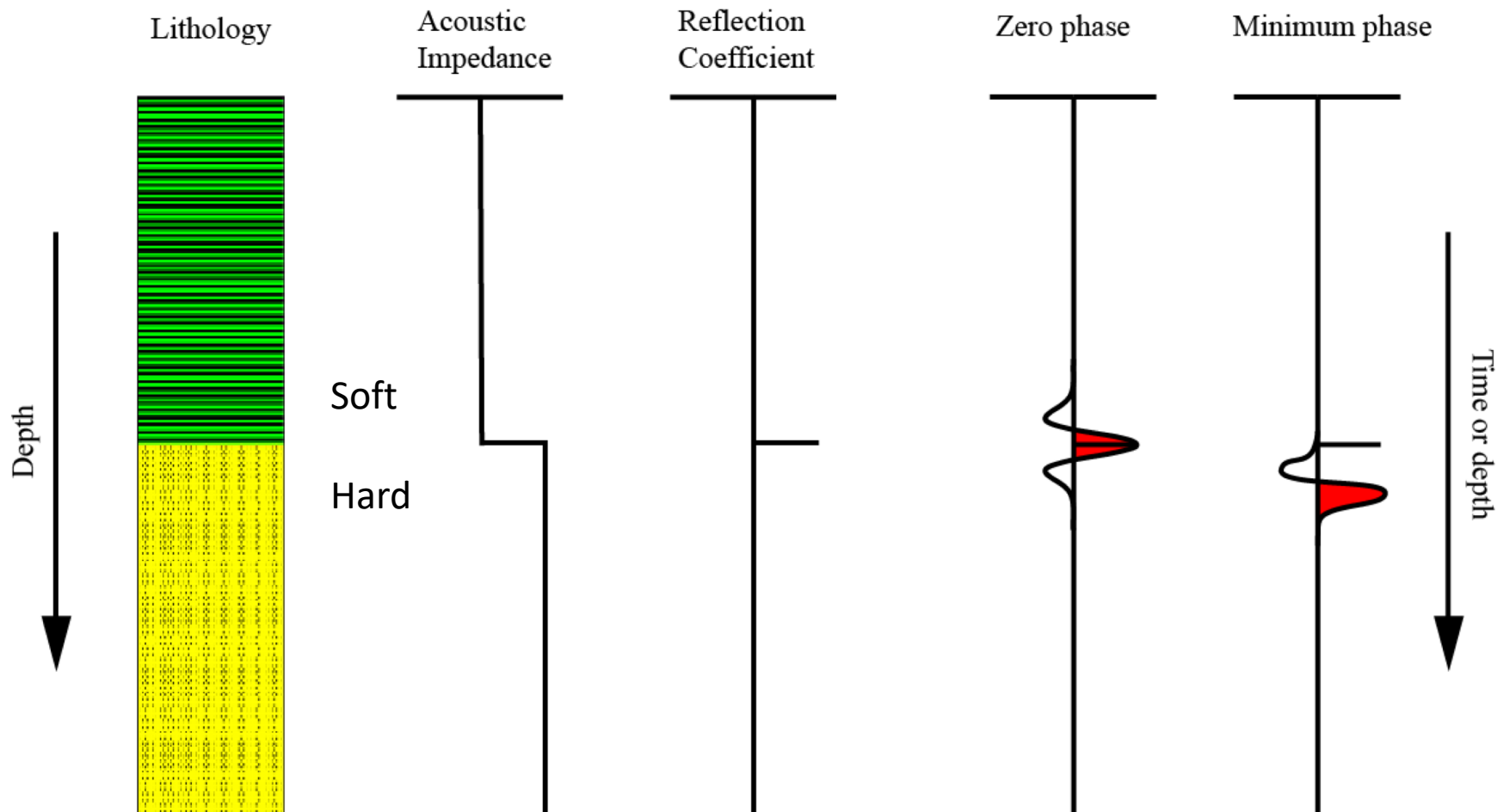
- Obtain a focussed image of reflections from geological contrasts in acoustic impedance with as high resolution as possible.
- Image reflectors at correct location and time and depth.
- Remove noise => No false reflectors.
- (Optional) Reflection amplitudes roughly proportional to those predicted by Zoeppritz' equation and vary correctly with incidence angle.



Seismic CMP gather

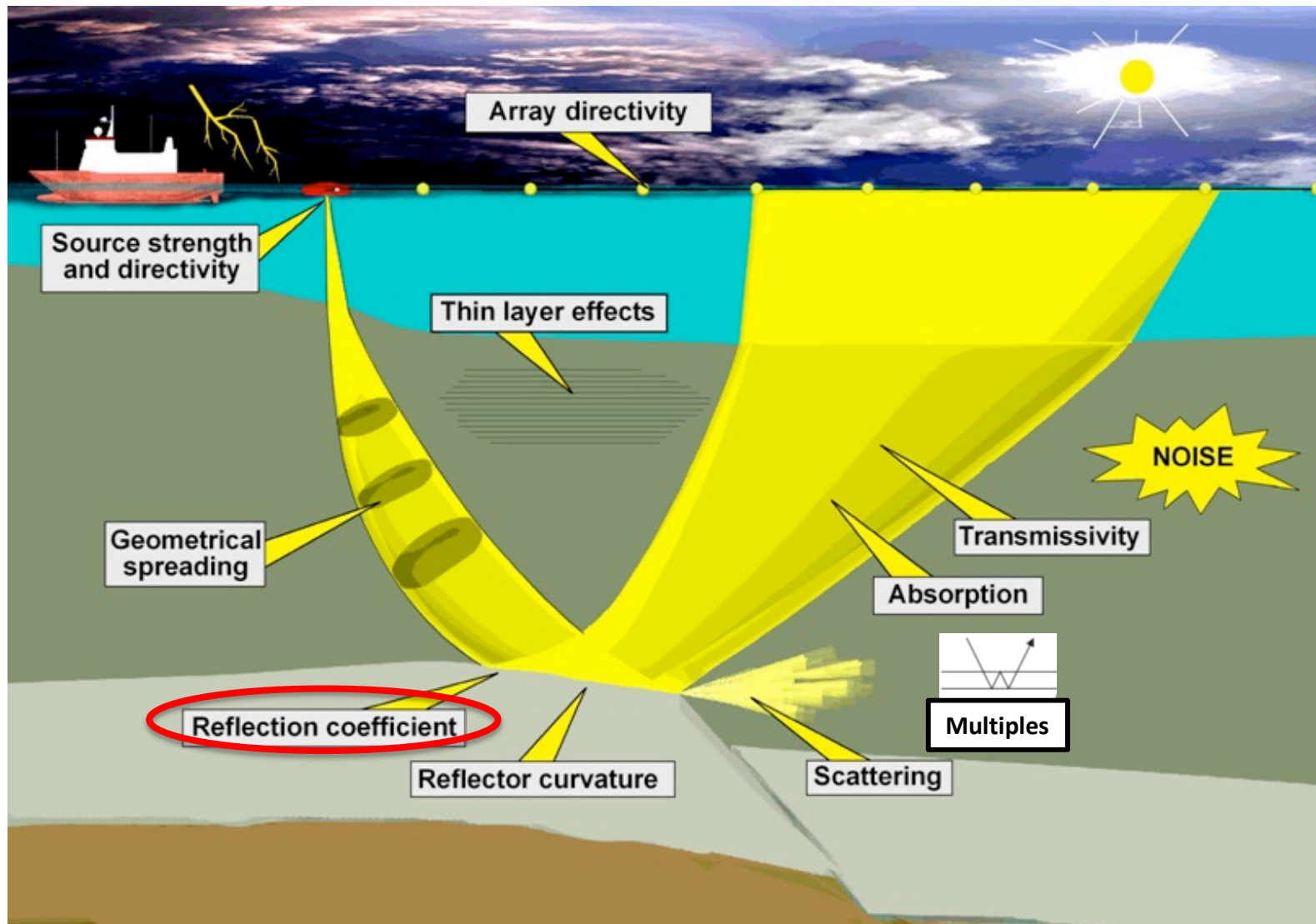


SEG normal* polarity standard



*SEG "Reverse" is simply the opposite of this polarity convention

Some effects on seismic amplitudes



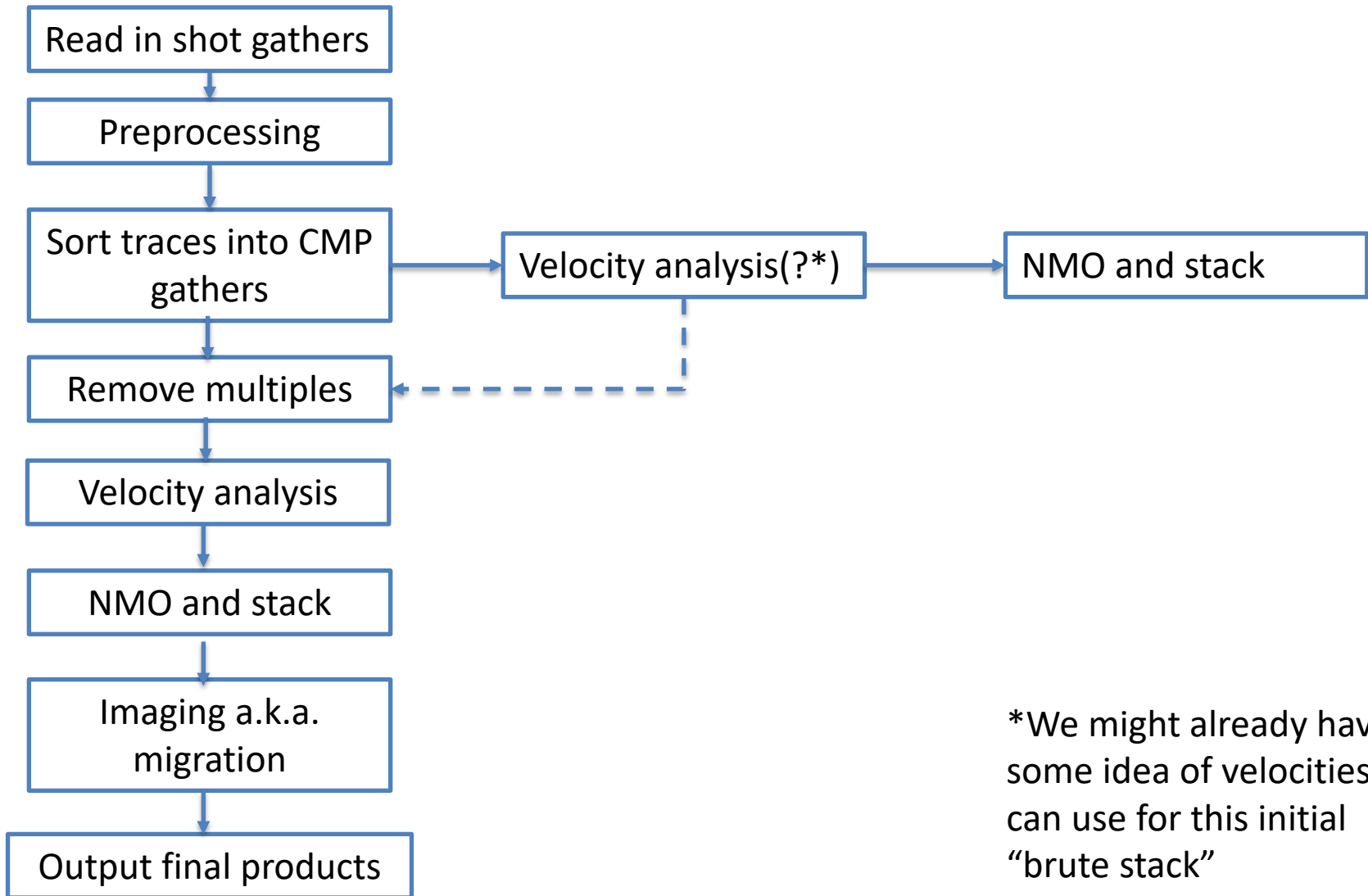
Signal and noise

Examples:

Coherent noise		Ambient noise	
Multiples and ghosts		Swell noise	
Direct wave		Rig or boat noise	
Refractions, guided waves, and diving/turning waves		Seismic interference	
Surface waves/ground roll		Rain or wind	
Mode conversions		Cable tug, bird noise etc.	
(Residual) bubble pulse		Earthquakes and microseismicity	

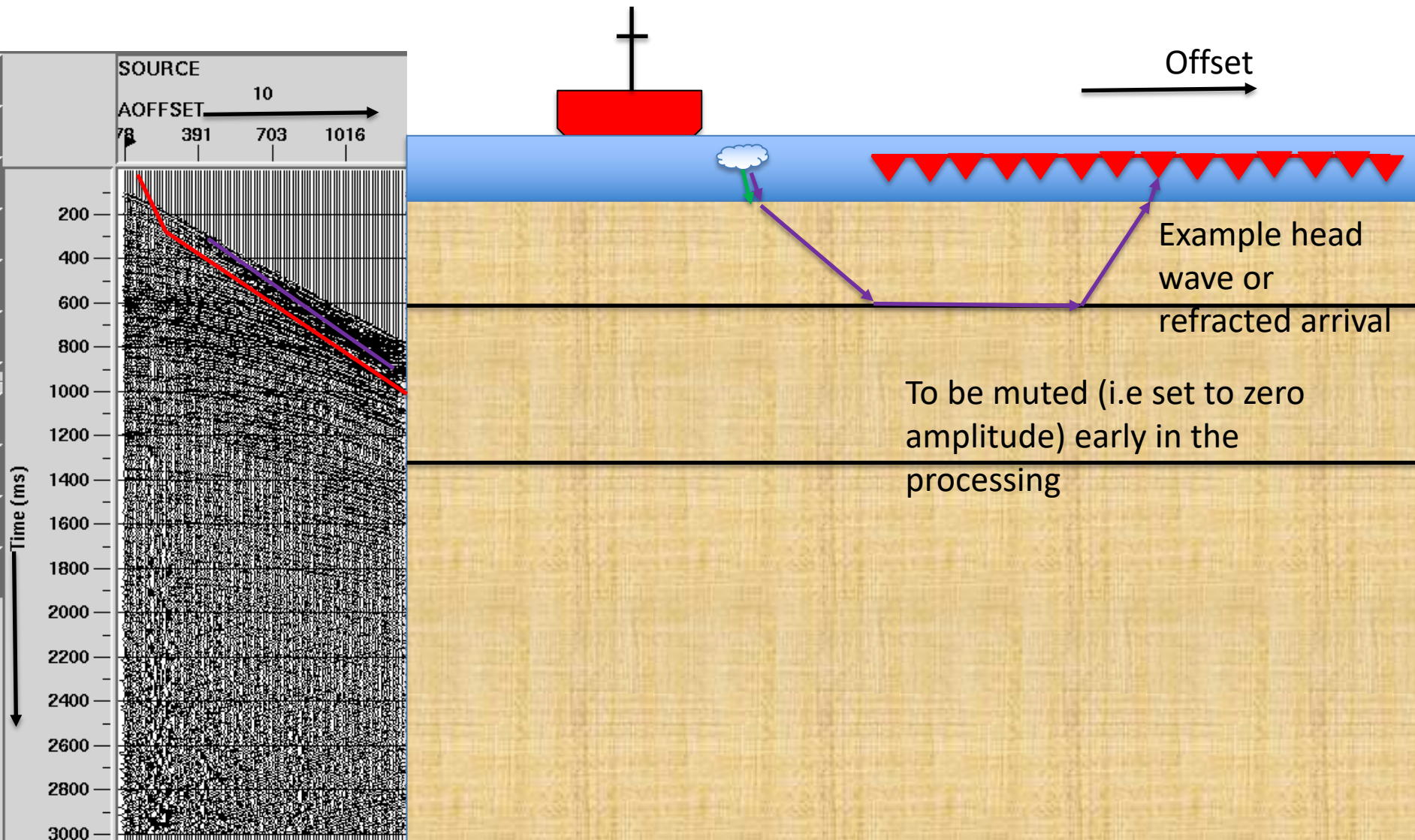
The signal is the parts of the seismic wavefield you can use to form an image or invert for subsurface properties. Usually, but not always, in reflection seismic, this is limited to primary reflections.

A very simple 2D processing flow illustrated

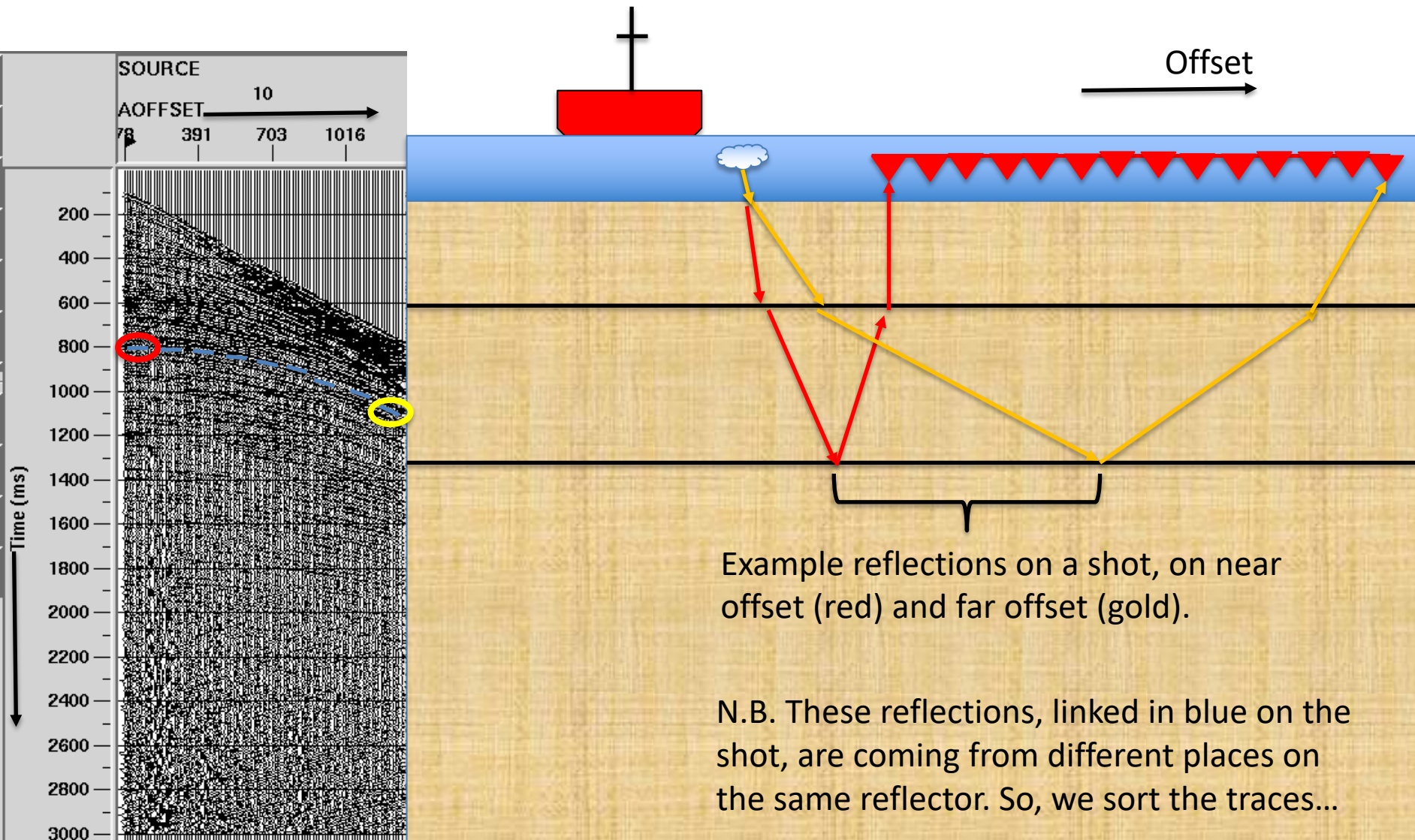


*We might already have some idea of velocities we can use for this initial “brute stack”

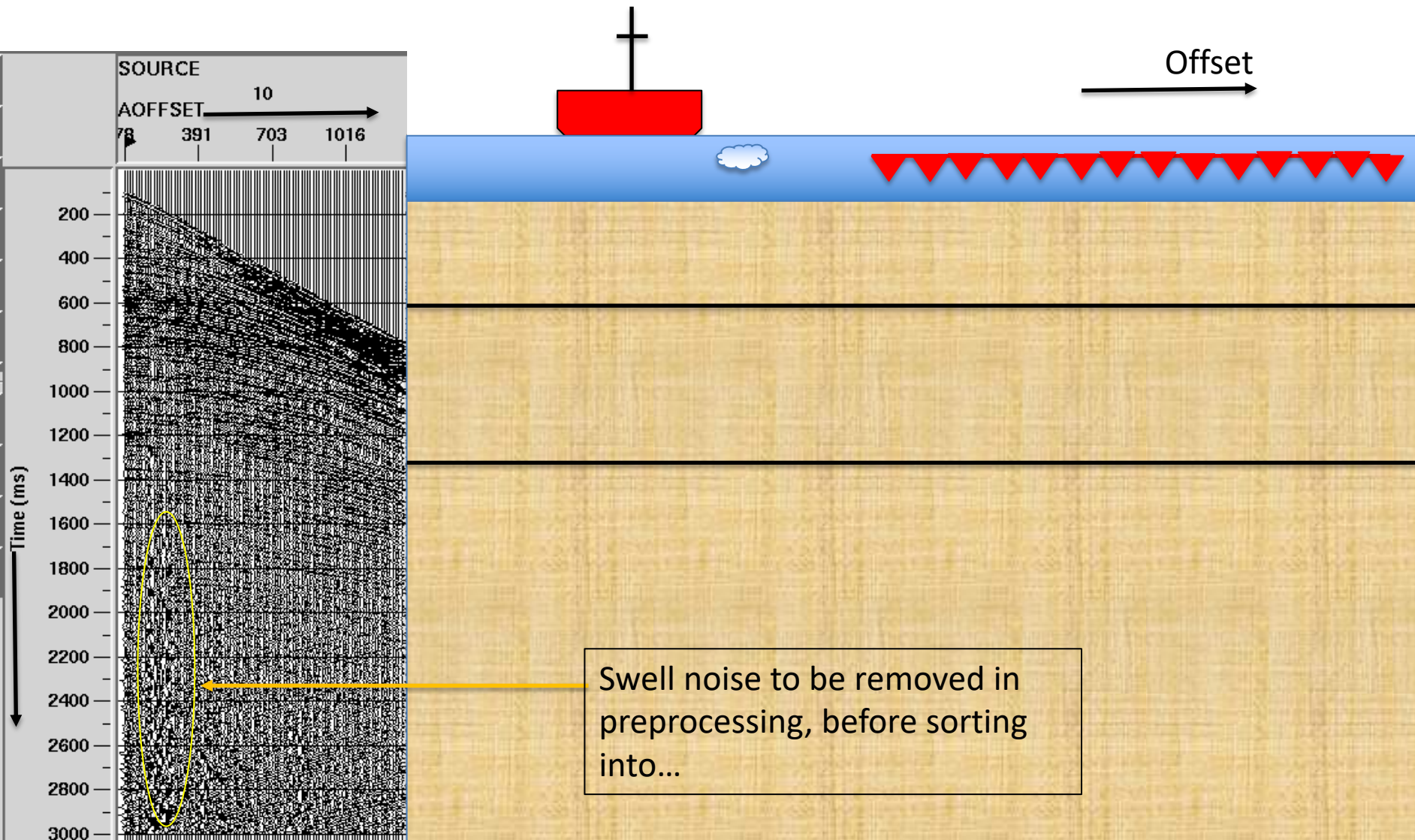
The Common Shot Gather ("A shot")



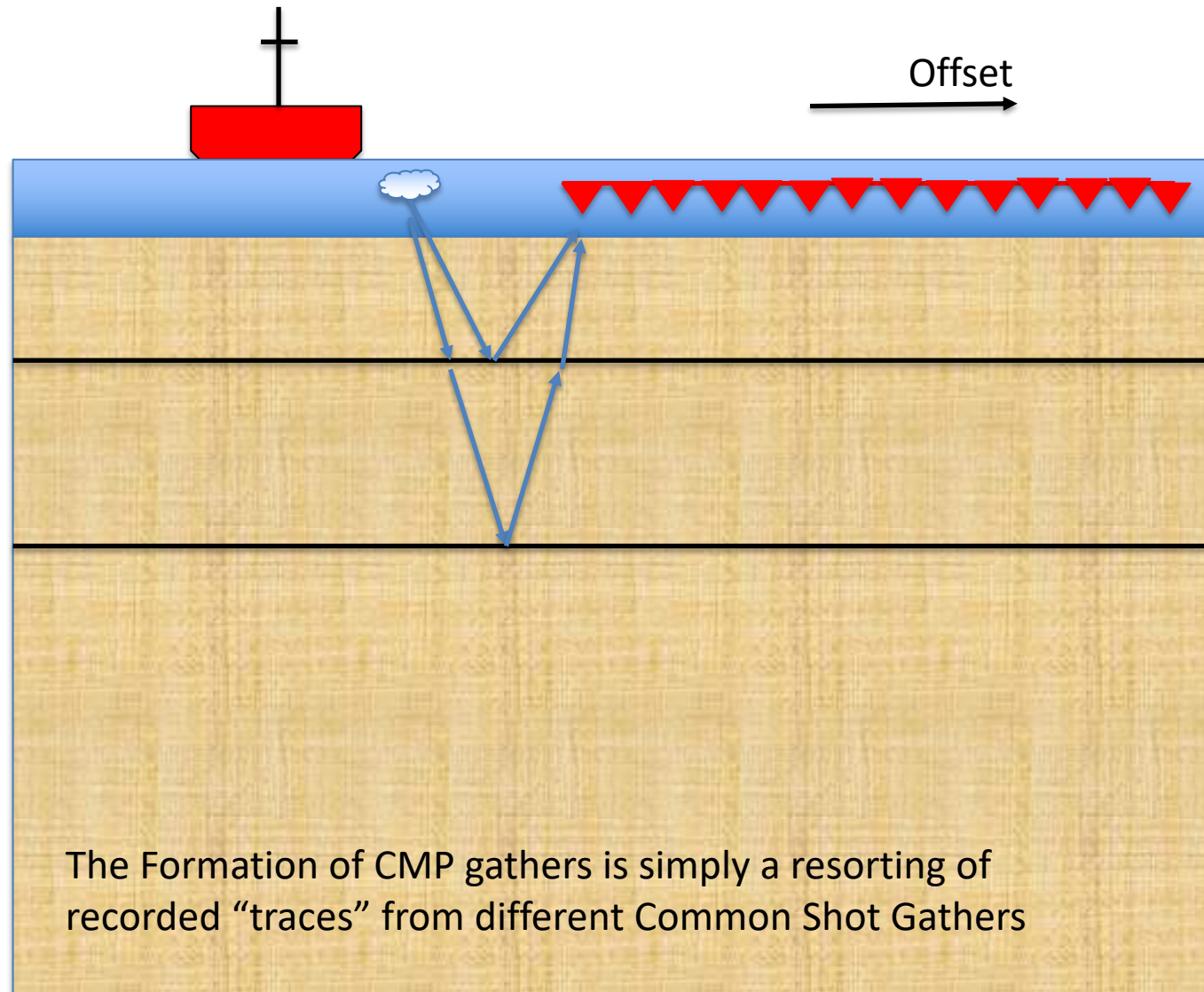
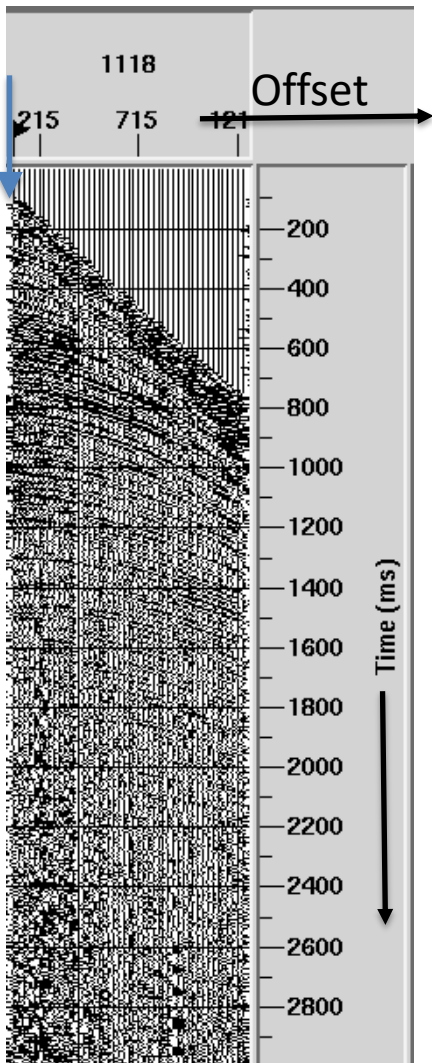
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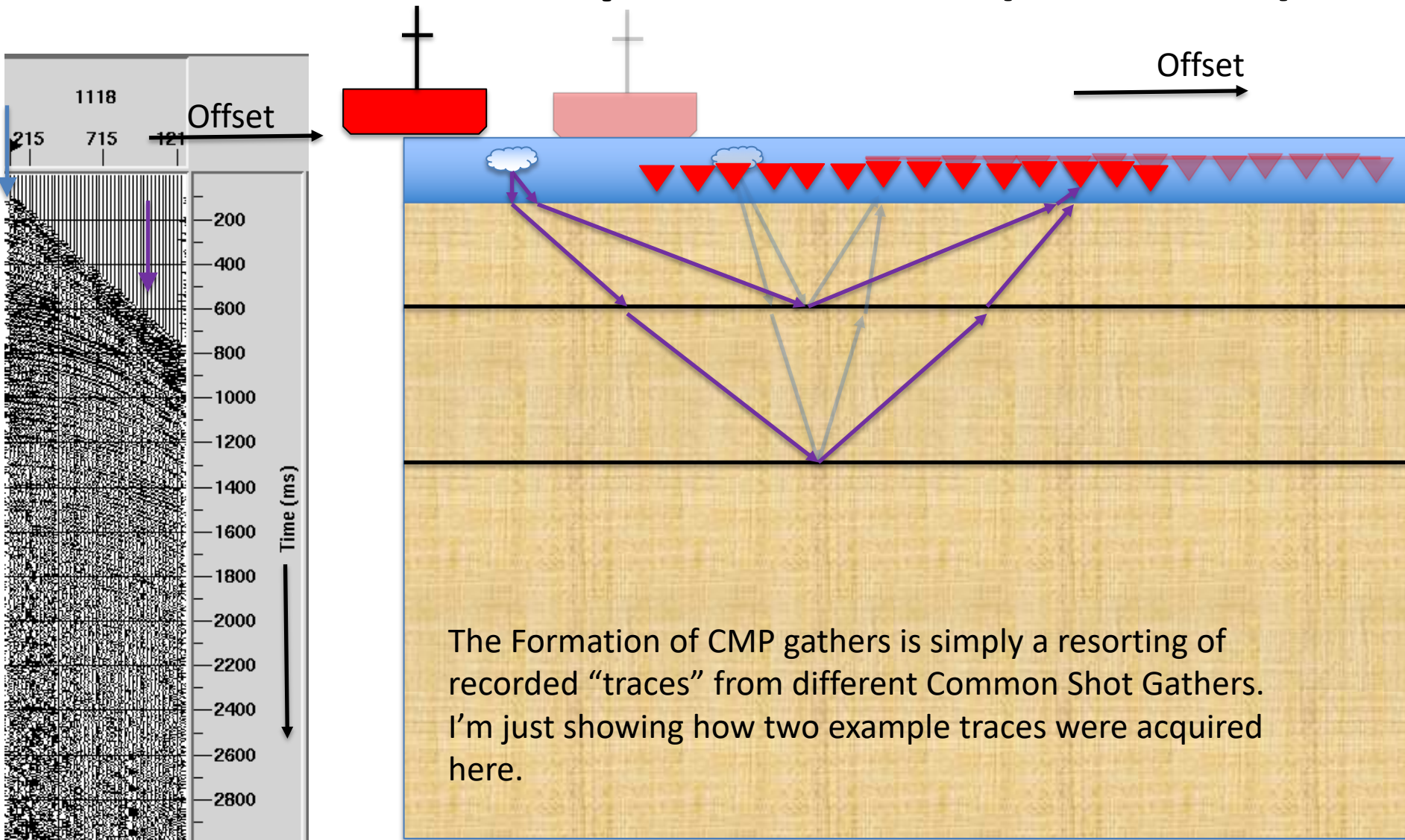
The Common Shot Gather ("A shot")



A Common Midpoint Gather ("A CMP")

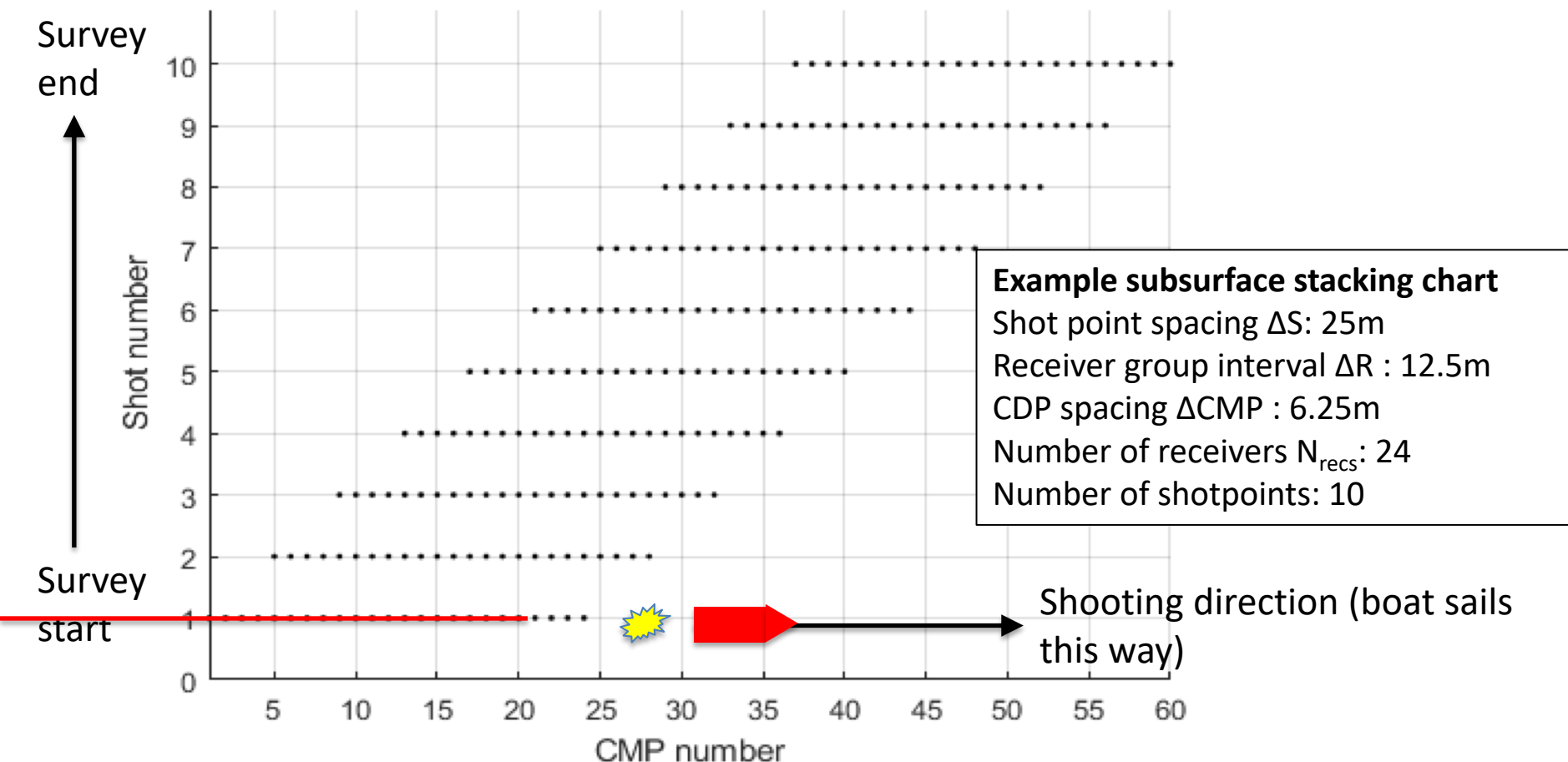


A Common Midpoint Gather ("A CMP")



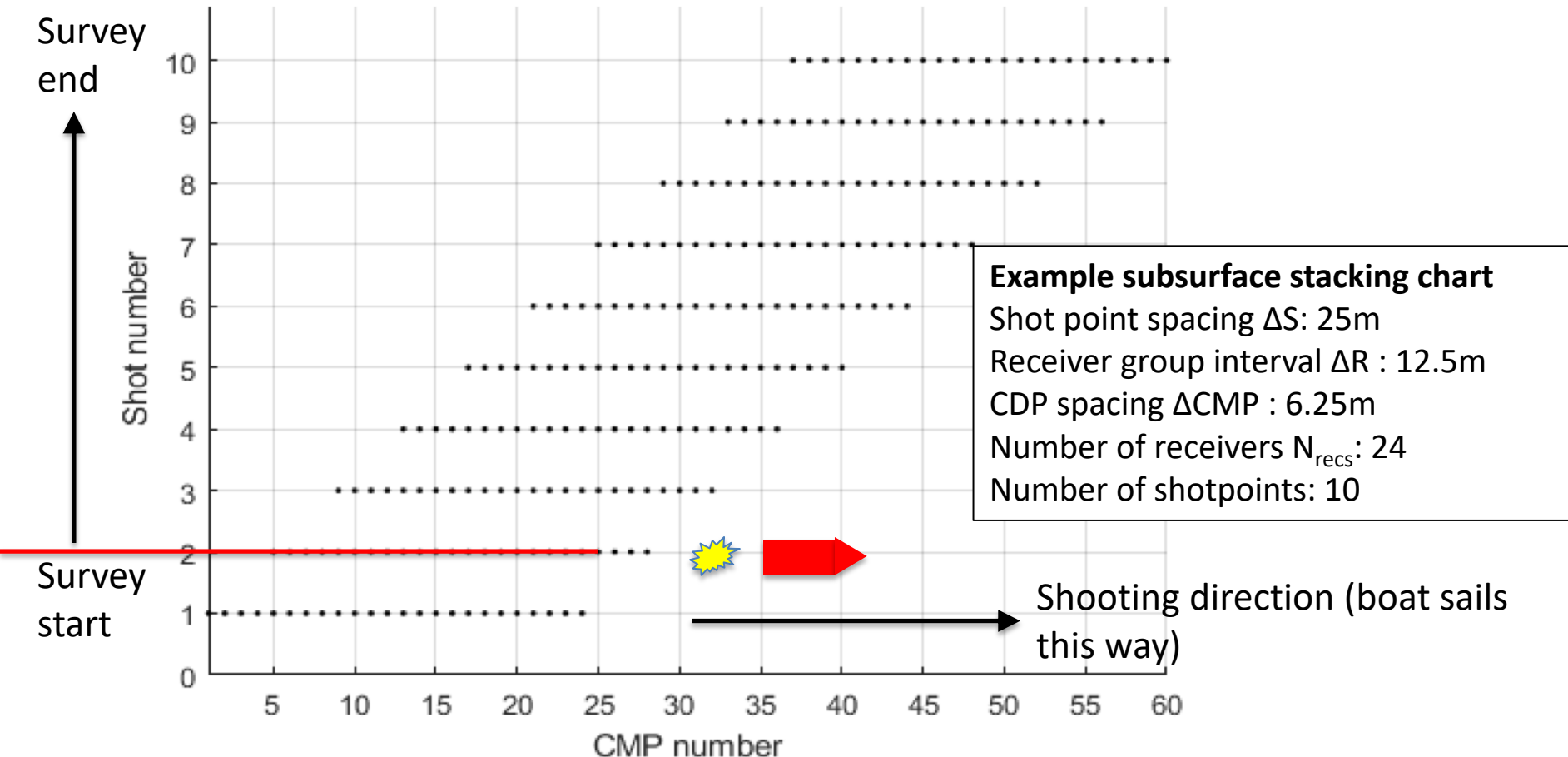
Example subsurface stacking chart

A subsurface stacking chart is a way of representing the subsurface coverage generated during acquisition, this one is for a short example line.



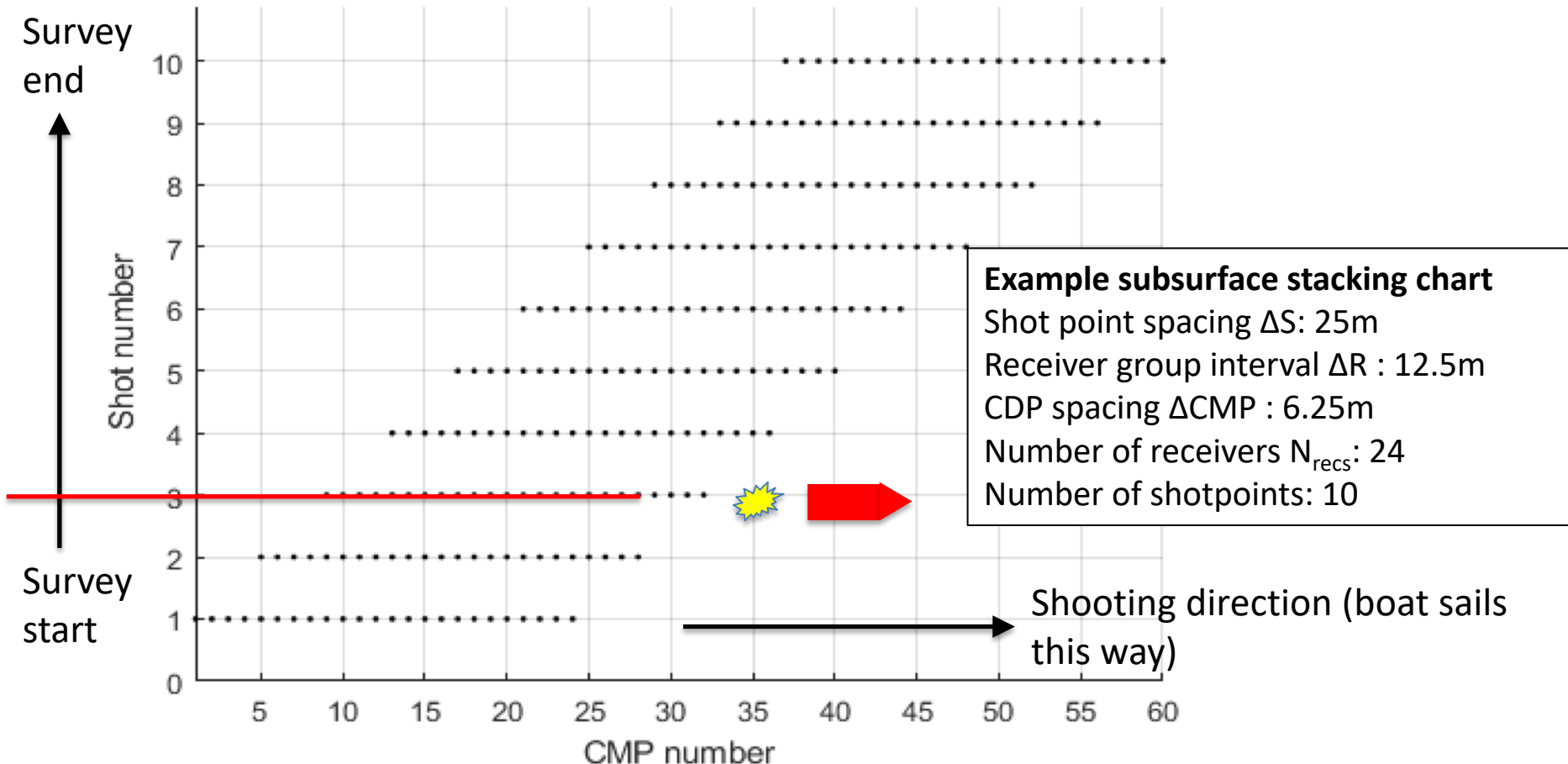
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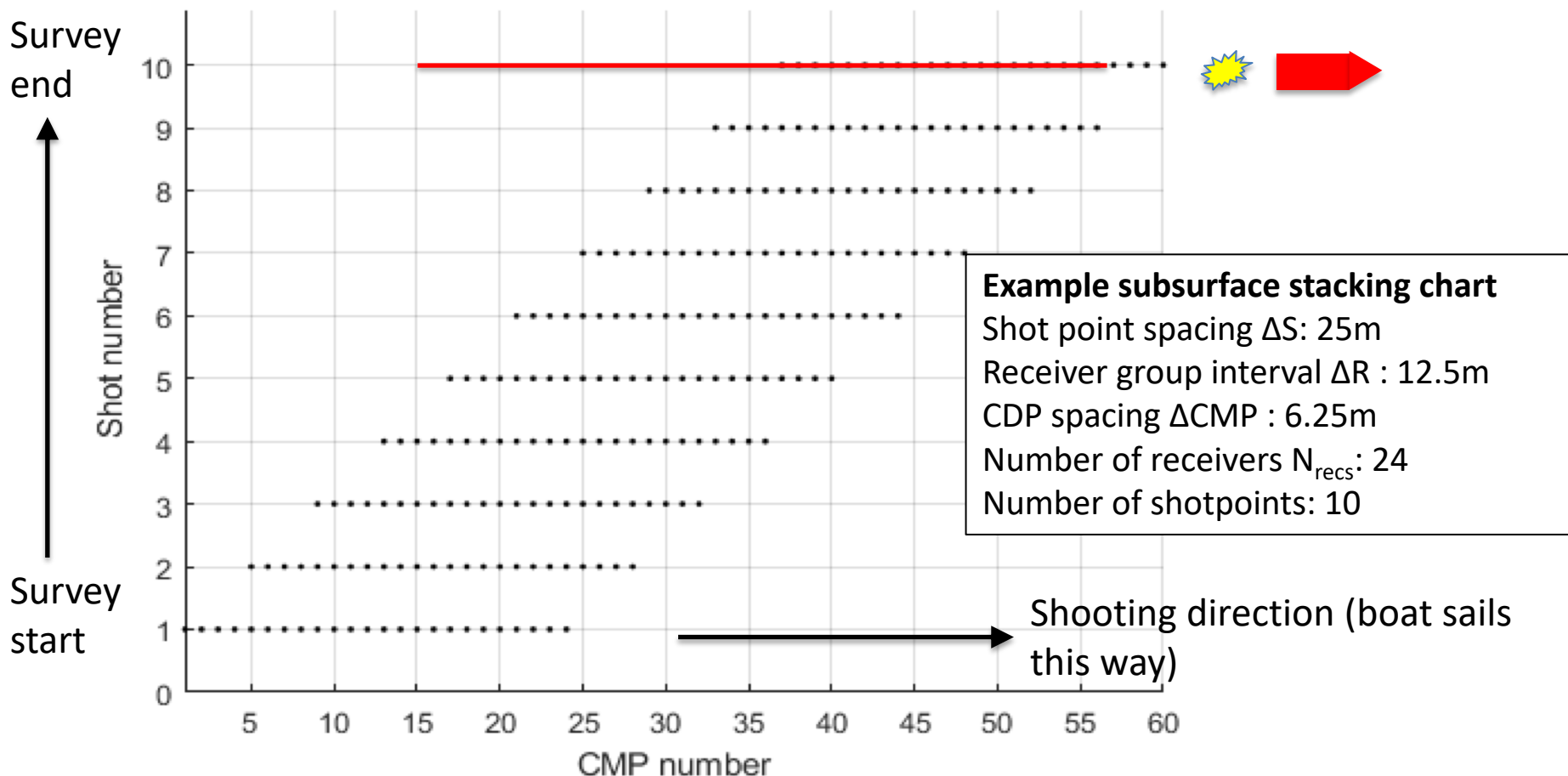
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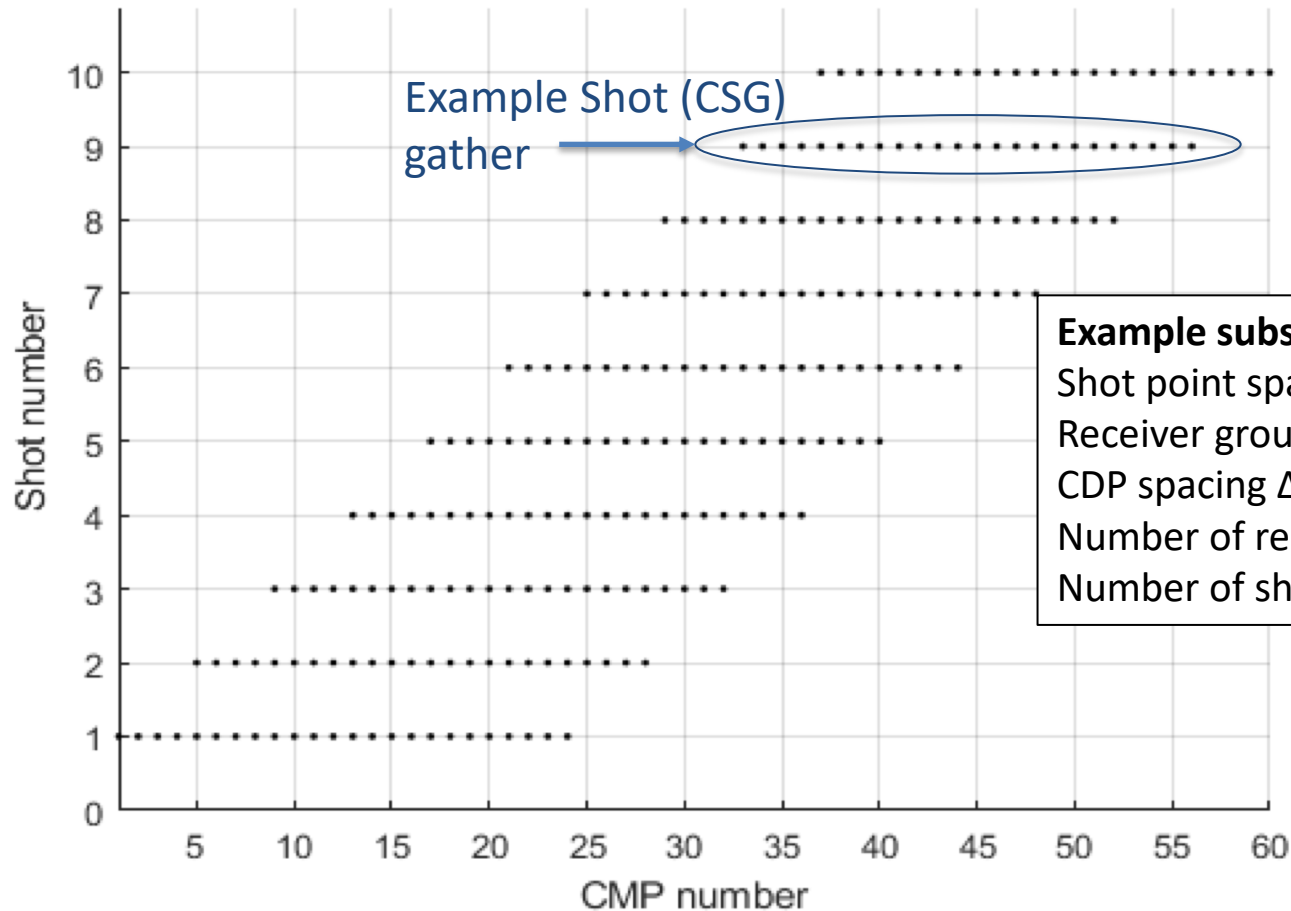


Example subsurface stacking chart

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Example subsurface stacking chart



Example subsurface stacking chart

Shot point spacing ΔS : 25m

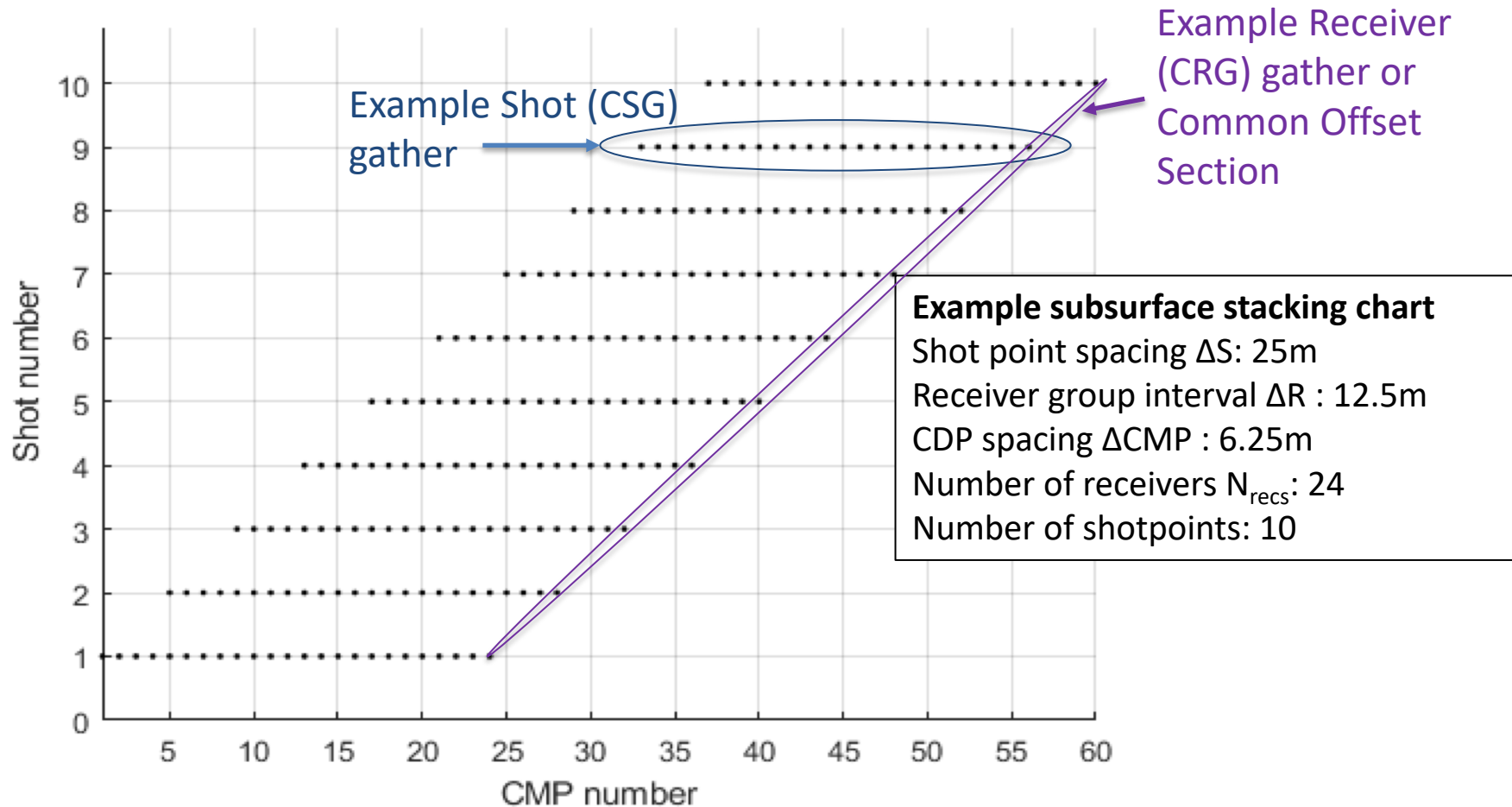
Receiver group interval ΔR : 12.5m

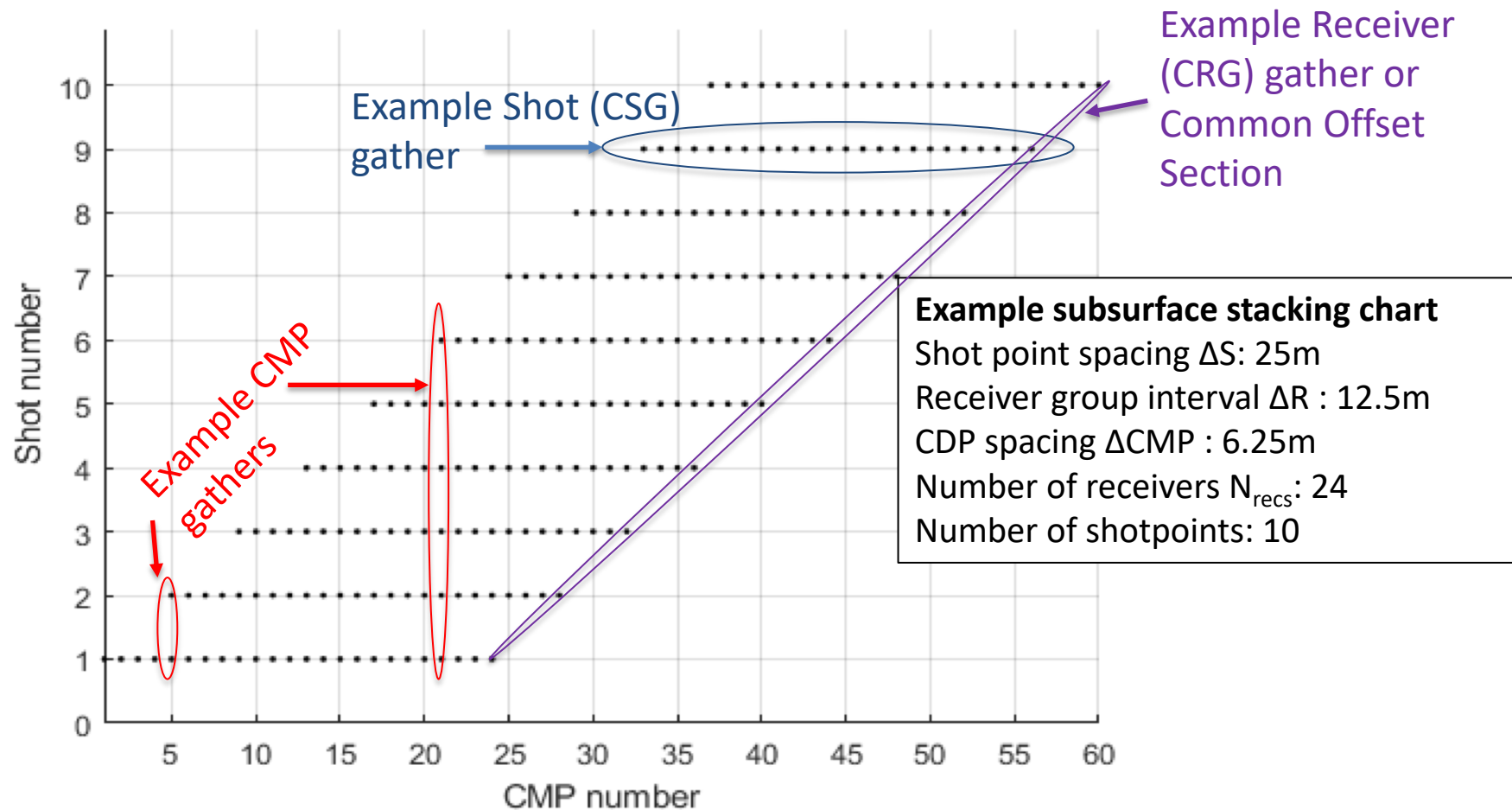
CDP spacing ΔCMP : 6.25m

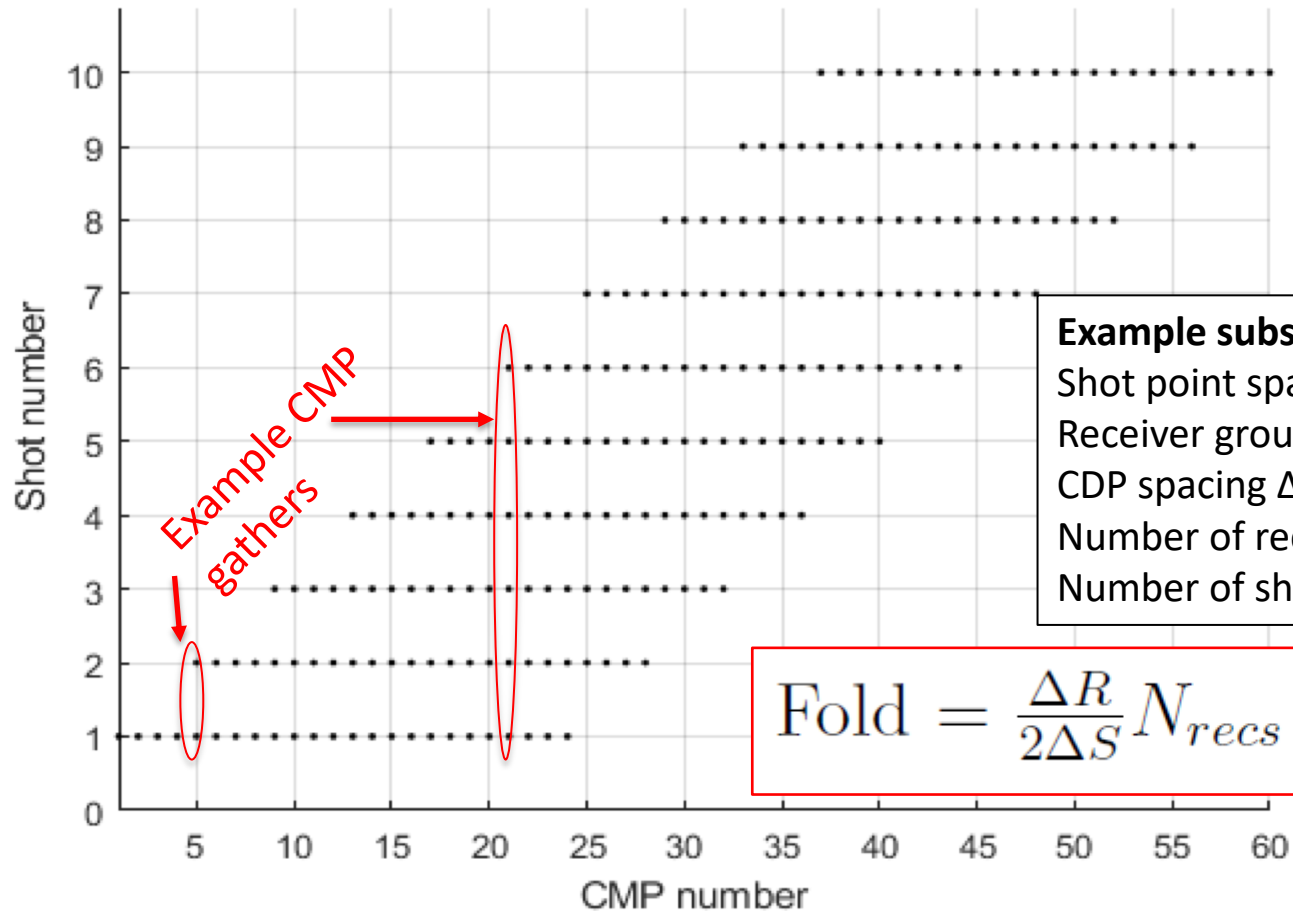
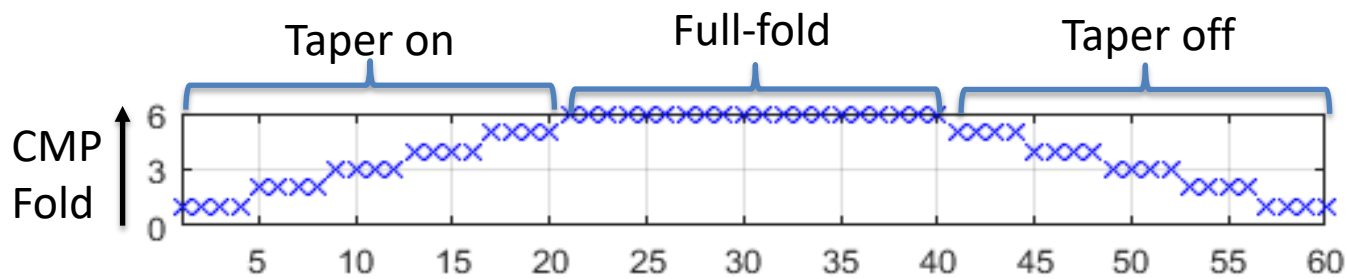
Number of receivers N_{recs} : 24

Number of shotpoints: 10

Example subsurface stacking chart







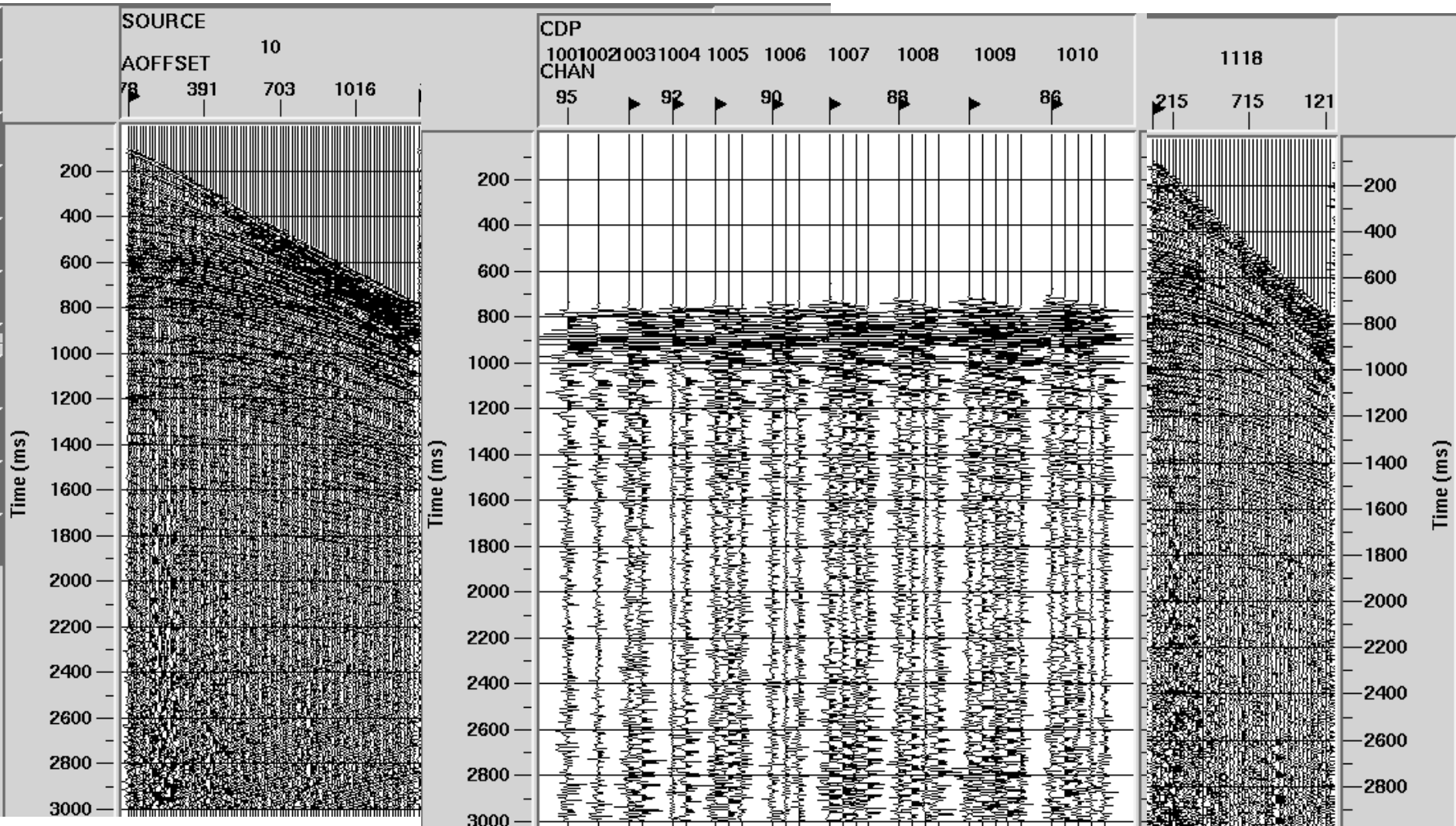
Example subsurface stacking chart
 Shot point spacing ΔS : 25m
 Receiver group interval ΔR : 12.5m
 CDP spacing ΔCMP : 6.25m
 Number of receivers N_{recs} : 24
 Number of shotpoints: 10

$$\text{Fold} = \frac{\Delta R}{2\Delta S} N_{recs} = \frac{\Delta CMP}{\Delta S} N_{recs}$$

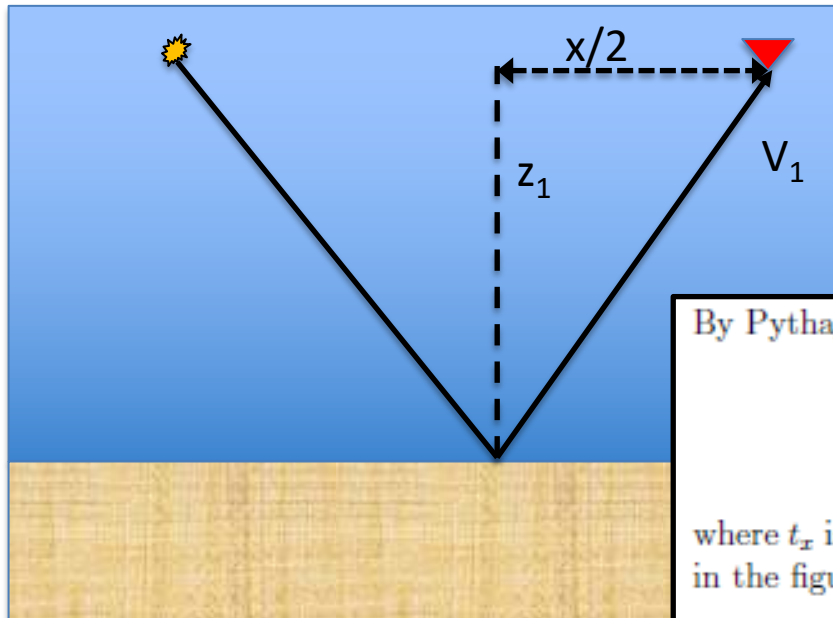
First Shot

First CMPs in line

Full fold CMP



Reflection from a single layer



Assume source and receivers at same level for now.

By Pythagoras' theorem:

$$t_x = 2\sqrt{\left(\frac{z_1}{V_1}\right)^2 + \left(\frac{x}{2V_1}\right)^2}$$

where t_x is the two way time from source to receiver and all other terms are defined in the figure. Squaring both sides gives:

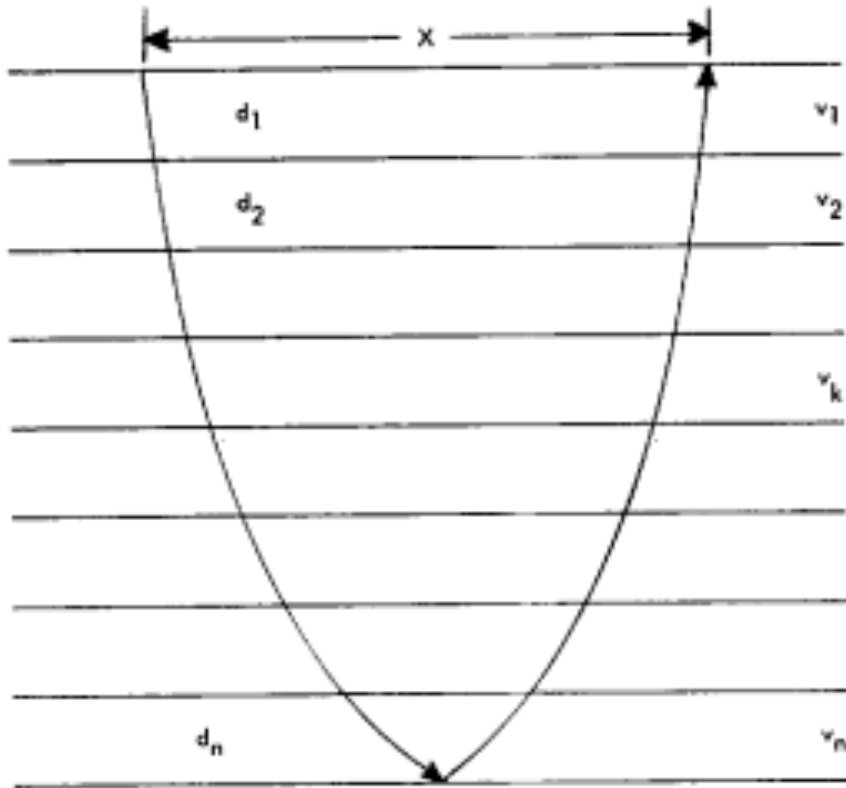
$$t_x^2 = 4\left(\frac{z_1^2}{V_1^2} + \frac{x^2}{4V_1^2}\right)$$

Substituting in vertical two-way time t_0 using $z_1 = \frac{v_1 t_0}{2}$ gives:

$$t_x^2 = t_0^2 + \frac{x^2}{V_1^2}$$

which is an exact expression for the “normal moveout”, or traveltimes dependence on source-receiver offset of a reflection through a single layer of constant isotropic velocity, V_1 .

Reflection from multiple layers



Taner and Koehler (1969) showed, using Fermat's principle, that for an isotropic horizontally layered medium the infinite travelttime series is:

$$t_{x,n}^2 = c_1 + c_2 x^2 + c_3 x^4 + c_4 x^6 + \dots$$

With the coefficients, c etc. depending on the layer thicknesses d and interval velocities v.

These days this is usually truncated at the x^2 term (see next slide) since beyond that we typically need to include anisotropy, and use alternative expressions.

Image courtesy: Taner and Koehler (1969)

NMO for the multi-layered case

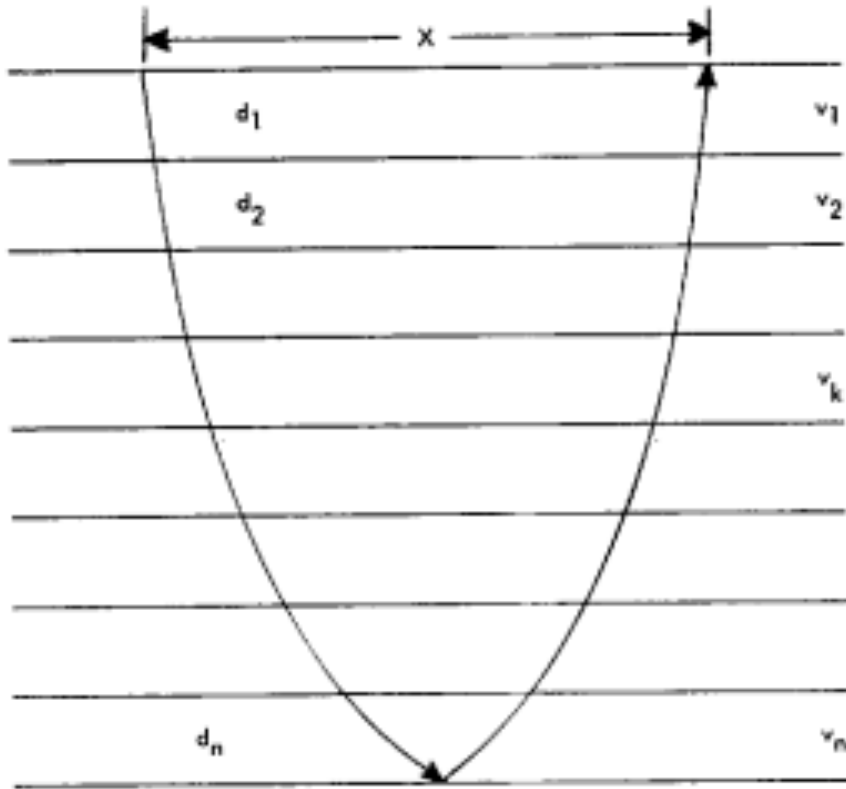


Image courtesy: *Taner and Koehler (1969)*

$$t^2 \approx t_0^2 + \frac{x^2}{V_{rms}^2}$$

The above holds for offset $x \ll$ reflector depth, z .

This velocity is used to correct for the hyperbolic increase in two-way-traveltime with increasing offset, the normal moveout (NMO) correction, before stacking.

RMS velocity, NMO, velocity, and stacking velocity, will be defined and discussed in a later lecture.

Velocity analysis

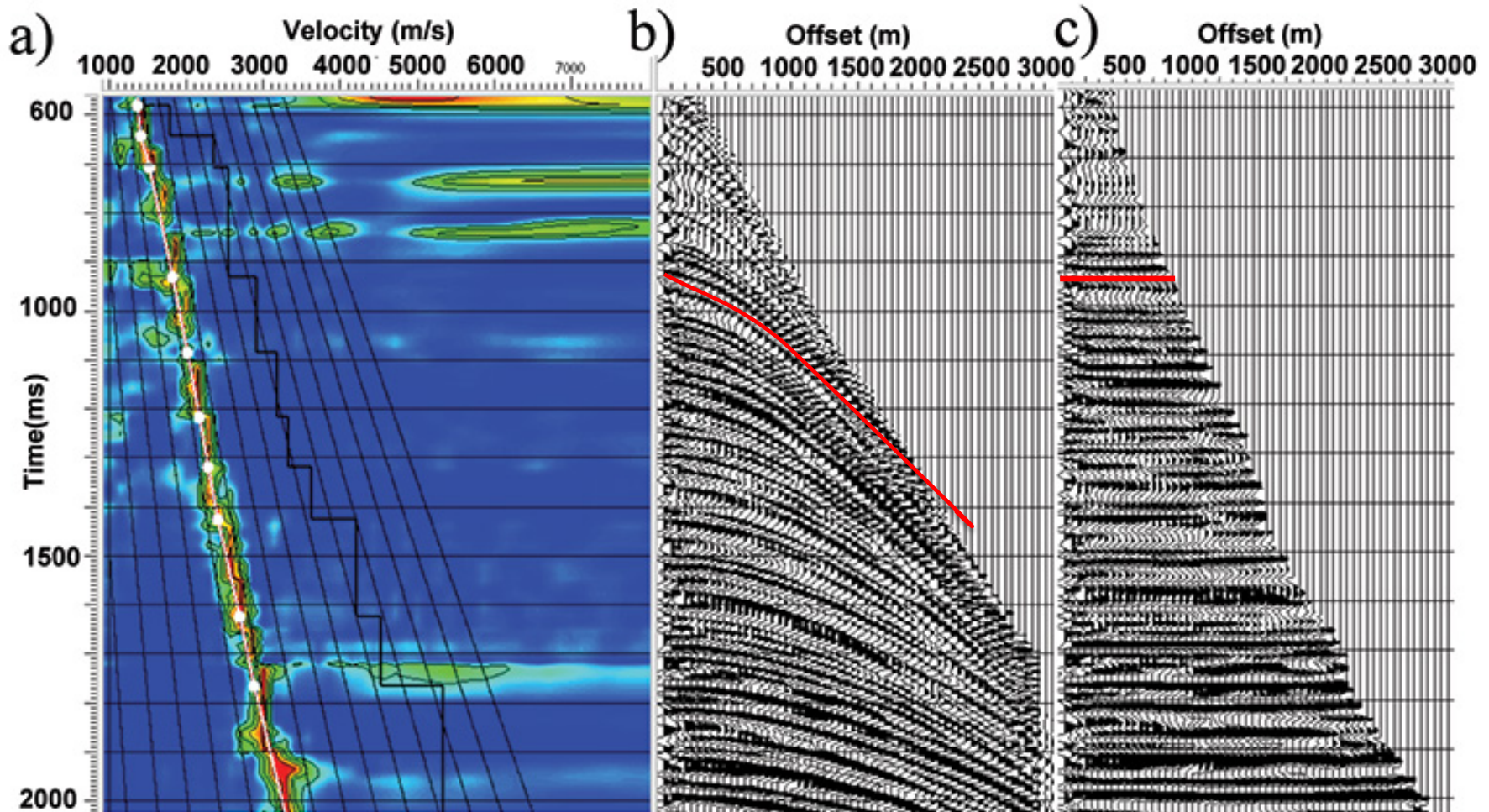


Figure courtesy: Sarkar (2013)

Normal MoveOut correction

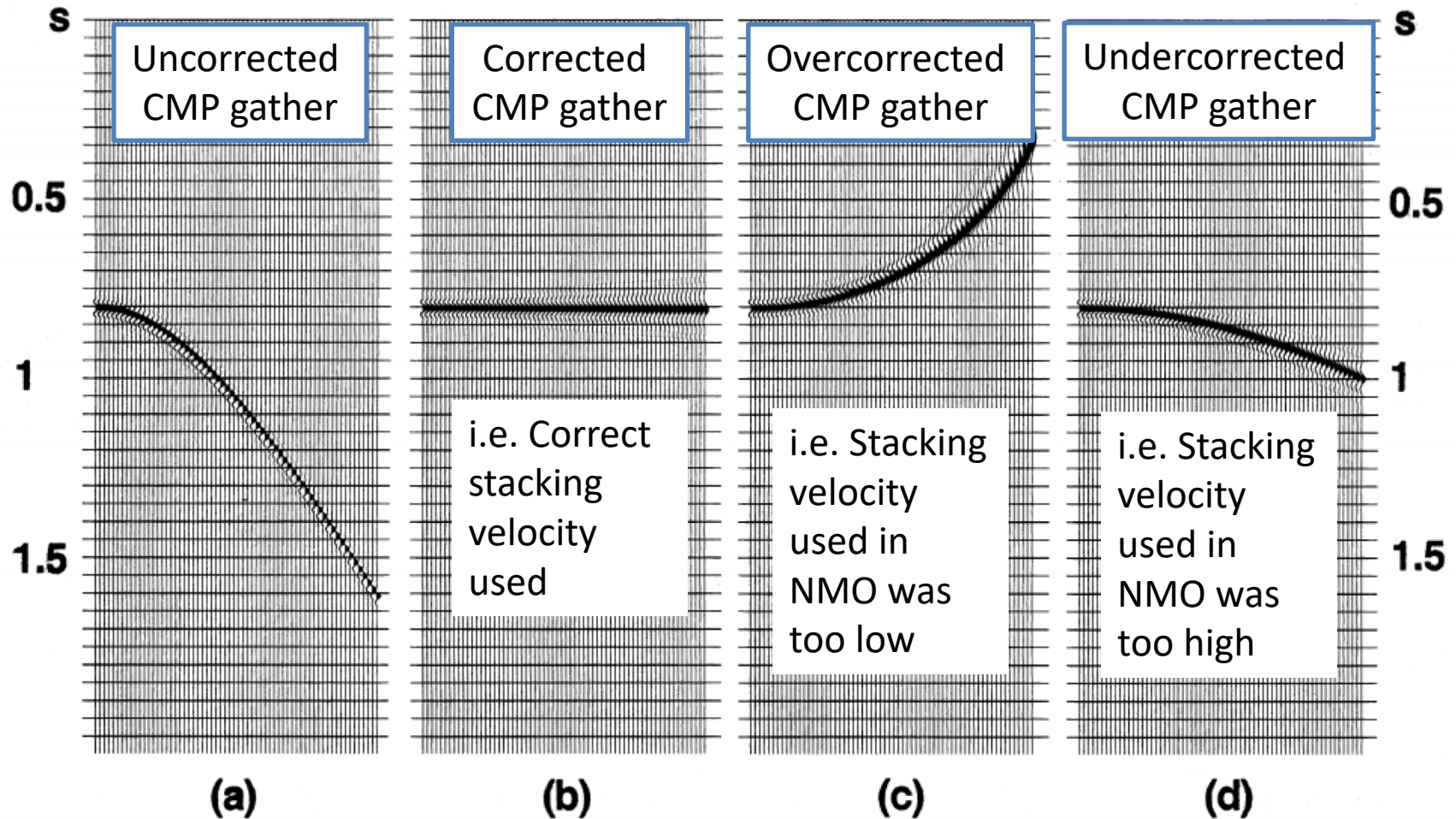
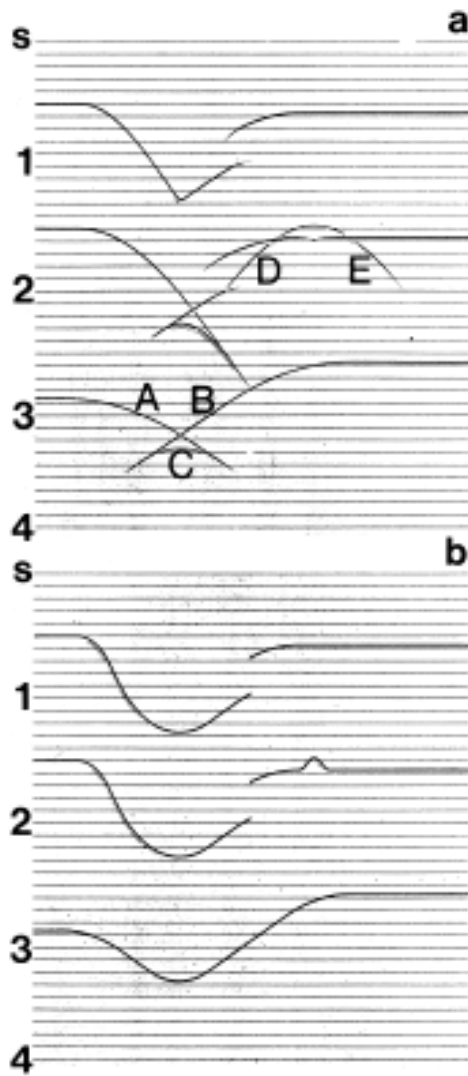


Figure courtesy: Yilmaz (2001)

Multiple Removal

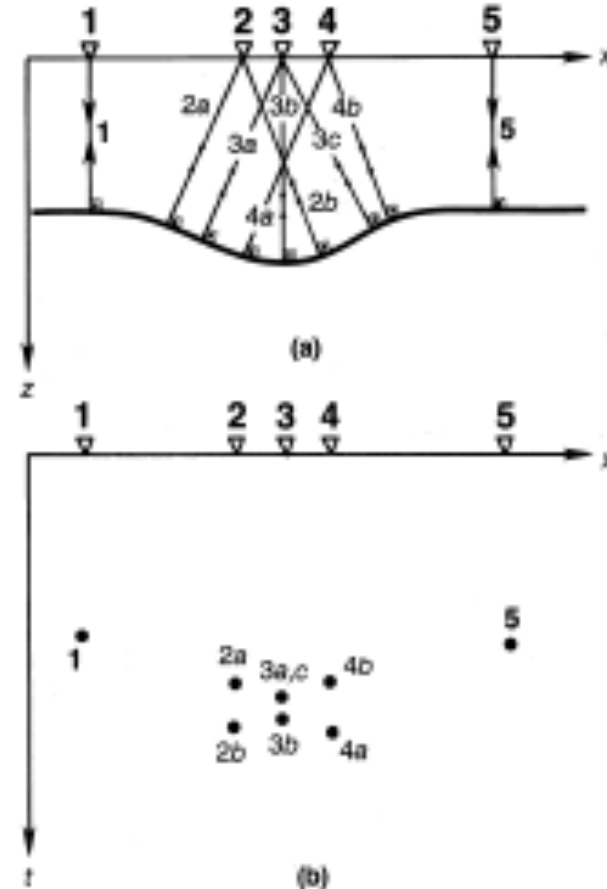
- Multiples are seismic reflection events that have reflected more than once between source and receiver.
- Multiple removal is important to avoid their potential misinterpretation by interpreters and velocity pickers as primaries.
- Ghosts are a special class of multiple.
- Multiples can be removed or attenuated based on:
 - Their predictability and/or periodicity.
 - Their lower velocities than primaries (exceptions to this are rare). These methods need velocity input.
- In the practicals, we will use deconvolution – which relies on periodicity of multiples, and F-K demultiple, which relies on their slower velocities.

Seismic Imaging



Synthetic seismograms over various synclines before (a) and after (b) migration.

Image courtesy: Yilmaz (2001)



Graphical explanation of why “bow-ties” arise on stacks at synclines before migration, see also deepest structure on the left. *Image courtesy: Yilmaz (2001)*

The End - Thank you for your attention!